Solving Equations with Variables on Both Sides

Collecting variable terms on one side, building the equation from a model, and recognizing special cases

The balance model behind every step

- An equation is a balanced scale. Whatever you do to one pan you must do to the other, or the balance is lost.
- **Step 1:** clear any brackets. **Step 2:** take the smaller variable term off *both* pans, so the variable survives on one side only. **Step 3:** undo the constant, then undo the coefficient.
- Every move undoes something, so use the *inverse* operation: a subtraction is cleared by adding.
- Check by substituting your answer back into the original equation.

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1. A balance scale is level. Each bag on it holds the same unknown number of blocks, and the loose blocks are all identical.

Left pan	**Right pan**
3 bags and 4 blocks	one bag and 12 blocks

Write the balance as an equation in x, where x is the number of blocks in one bag, and solve it.

Answer: ________________

2. Solve: $5x = 2x + 9.$

Answer: ________________

3. Solve: $7x - 5 = 4x + 10.$

Answer: ________________

4. A gym charges a joining fee of 20 dollars plus 3 dollars for each class. A rival gym charges a joining fee of 8 dollars plus 5 dollars for each class. After how many classes do the two gyms cost the same? Write an equation in n and solve it.

Answer: ________________

5. Solve: $4(x - 2) = 2x + 6$.

Answer: ________________

6. Consider the equation $2(x + 3) = 2x + 6$.

(a) Clear the bracket and write what each side becomes. Answer: ________

(b) Explain how many solutions the equation has, and why.

7. Solve: $3(2x + 1) = 5x - 4$.

Answer: ________________

8. Two candles are lit at the same moment. One starts at 24 centimetres and burns down 3 centimetres each hour. The other starts at 16 centimetres and burns down one centimetre each hour. After how long are the two candles the same height? Write an equation in t and solve it.

Answer: ________________ hours

9. Consider the equation $5x + 3 = 5x - 2.$

(a) Collect the variable terms and write what is left. Answer: _________

(b) Explain how many solutions the equation has, and why.

10. **Challenge.** Elena solves $7x - 6 = 2x + 9$ like this:

$$7x - 6 = 2x + 9 \longrightarrow 5x - 6 = 9 \longrightarrow 5x = 3 \longrightarrow x = \frac{3}{5}$$

(a) Solve the equation correctly. Answer: ________

(b) Explain which step went wrong and what rule Elena broke.